Atlassian April 2022 Mega Outage Technical Breakdown

Script Parameter Confusion, Soft-Delete Cascades & The 14-Day Restoration Bottleneck

POST-MORTEM

CLOUD RELIABILITY

1. Incident Overview & The 14-Day Paradox

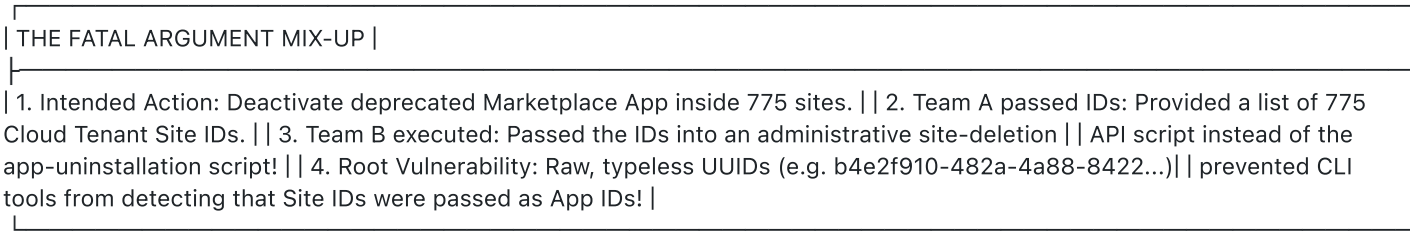
On April 5, 2022, 775 enterprise businesses suddenly lost access to Jira, Confluence, Jira Service Management, and Opsgenie. Unlike typical cloud outages caused by hardware failures or routing loops that resolve within hours, this outage persisted for **14 consecutive days (336 hours)**.

The Engineering Paradox

Zero customer data was lost or corrupted. All relational records and attachments remained safely encrypted in AWS EBS volumes and Aurora RDS snapshots. Yet, 775 global enterprise sites were completely offline for two weeks due to the total absence of bulk tenant restoration tooling.

2. The Fatal Miscommunication: Script Argument Confusion

Atlassian acquired *Insight - Asset Management* and bundled it natively into Jira Service Management. To clean up redundant Marketplace standalone installations, engineering scheduled a deactivation script targeting 775 customer tenants that possessed both versions:



3. The Automated Cascading Soft-Delete Event Storm

Cloud architectures prioritize automated, fast teardown for privacy and cost control. When the script invoked `mark_site_deleted(site_id)`, an automated cascade triggered across all microservices:

Routing & Ingress Layer

Global API Gateways instantly dropped DNS virtual host mappings. All incoming web and API traffic for 775 subdomains returned HTTP 404 / 410 Gone.

Identity & Atlassian Access

Central provisioning revoked SAML SSO tokens, decoupled user directory syncs (Okta/Azure AD), and invalidated active OAuth bearer sessions.

Jira & Confluence Datastores

Tenant database schemas were detached from connection pools and marked tombstoned; Elasticsearch/Lucene search indices were dropped.

Object Storage (S3)

Customer file attachments and avatars were unlinked and transitioned to orphaned lifecycle state. The entire teardown took less than 3 minutes.

4. The Asymmetry of Operational Automation

The critical flaw exposed by the incident was an extreme imbalance between deletion and recovery tooling:

- **Automated Deconstruction:** High-throughput, asynchronous, event-driven pipelines engineered to purge tenant resources in seconds.
- **Manual Reconstruction:** Zero bulk undelete capability. Site deactivation was treated as an irreversible business action; restoration was a rare, manual support playbook executing over hours.

5. Multi-Tenant RDS Sharding: The 336-Hour Bottleneck

Why couldn't Atlassian simply restore an AWS Aurora snapshot from before the script execution?

> **The Shared Multi-Tenant Database Conundrum**
> Atlassian provisions hundreds of enterprise customer tenants within the same shared Amazon Aurora database cluster, differentiated logically by schemas and tenant IDs. Restoring a point-in-time snapshot over a live shard would overwrite hours of legitimate transactions for thousands of unaffected, active tenants!

Manual Multi-Step Recovery Protocol (Repeated for Each of the 775 Tenants): 1. Spin up an isolated, temporary Aurora RDS cluster from historical EBS snapshot. 2. Query snapshot to isolate target tenant's relational tables, custom fields & foreign keys. 3. Export tenant data dumps and sanitize against latest schema migration deltas. 4. Inject extracted schema into live production shard without holding long-lived write locks. 5. Re-establish SAML SSO mappings in Atlassian Access & regenerate OAuth security keys. 6. Rebuild Lucene/Elasticsearch indices from scratch and run manual end-to-end customer audits. Execution Time per Customer Site: 2 to 8 hours of senior database engineering labor!

6. Comparative Matrix: Fragile vs. Resilient Operations

Architectural Vector	Pre-Incident Fragile Operations	Hardened Resilient Operations
Identifier Safety	Raw UUIDv4 strings (identical across all domain entities)	**Strongly typed, prefixed IDs (`site_...` vs `addon_...`)**
Blast Radius Limits	Unbounded array inputs (1 to 100,000 entities per script)	**Strict limits: Max 5/batch, mandatory canary pauses**
Execution Governance	Single-operator terminal execution	**Multi-party 4-eyes approval with cryptographically signed tokens**
Restoration Capability	Manual, serialized snapshot extraction scripts	**Automated multi-service bulk tenant reconstitution API**
Disaster Recovery Drills	Infrastructure-level AZ/Region failover only	**Logical tenant-level corruption and deletion chaos drills**

7. Systemic Root Cause Analysis

- **Absence of Type-Safe Domain Identifiers:** Passing a list of Site IDs into an App ID parameter was accepted silently by CLI scripts because both were generic strings.
- **Unbounded Operational Blast Radius:** Administrative endpoints lacked circuit breakers to detect anomalous bulk operations (e.g., deactivating >10 enterprise tenants in a single call).
- **Neglect of Reverse-Path Tooling:** Massive investment in teardown pipelines for GDPR compliance without reciprocal investment in instant restoration pipelines created an operational asymmetry.

8. Architectural Reforms Implemented Post-Incident

1. Strongly Typed Prefixed IDs

All internal APIs enforce semantic prefixes (`site_live_...`, `addon_insight_...`). Mismatched prefixes fail immediately at API gateways with HTTP 422.

2. Blast Radius Circuit Breakers

Administrative actions cap batch sizes to 5 tenants with mandatory canary evaluations. Deletion rates exceeding baseline automatically trip emergency circuit breakers.

3. Multi-Party Authorization (4-Eyes)

High-risk production operations require independent sign-offs from two authorized engineers using time-bounded cryptographic tokens.

4. Automated Reconstitution API

Tenant soft-deletes preserve encrypted metadata pointers for 30 days. An automated orchestration API reconstitutes cross-service tenant states in under 5 minutes.

9. Key Architectural Takeaways for Distributed Systems

Core Reliability Principles Distilled from the Atlassian Outage

- **Deletion Without Tested Rollback Is Technical Debt:** If an automated system can tear down an environment in minutes, an automated system must exist to restore it in minutes. Untested recovery procedures fail when needed most.
- **Guard Against Human Error with Blast Radius Limits:** Administrative tools must assume bad inputs. Strict batch caps, rate limiters, and canary phases prevent a single mistaken argument from becoming an enterprise disaster.
- **Multi-Tenant Storage Demands Logical Extraction Tooling:** Shared physical shards render full database snapshot restoration useless for single-tenant recovery. Build logical tenant snapshot and streaming replication tools upfront.
- **Encode Domain Meaning in Identifiers:** Raw UUIDs invite catastrophic ambiguity. Type-prefixed IDs catch parameter bugs at runtime before destructive mutations execute.